

CML — Epidemiology, Phases & Response Milestones



1. 15% of leukemias

WHAT YOU SEE

A wall monitor reading 15%.

WHAT IT ENCODES

CML accounts for ~15% of all adult leukemias. It is a myeloproliferative neoplasm driven by BCR::ABL1. Incidence ~1–2 per 100,000/year, rising with age.

BOARD TRAP

CML is the classic MPN with the Philadelphia chromosome; do not confuse its 15% share with CLL, which is the most common adult leukemia overall in the West.

2. Age 55–65

WHAT YOU SEE

Monitor reading 55–65.

WHAT IT ENCODES

Median age at diagnosis is ~55–65 years. CML can occur at any age but is uncommon in children. Often discovered incidentally on a CBC showing leukocytosis with left shift and basophilia.

BOARD TRAP

A young patient with massive splenomegaly and a high WBC still fits CML — age range is a median, not a rule-out.

3. 1.6:1 male predominance

WHAT YOU SEE

Gender symbols reading 1.6:1.

WHAT IT ENCODES

Slight male predominance (~1.4–1.6:1). The only proven environmental risk factor is ionizing radiation exposure.

BOARD TRAP

Benzene and chemo cause AML/MDS, not classically CML — radiation is the CML-linked exposure.

4. Pre-TKI 3–6 yr survival

WHAT YOU SEE

Skull icons labeled PRE-TKI, 3–6 yr OS, 10–20% 10-yr.

WHAT IT ENCODES

Before imatinib, median survival was only 3–6 years with progression to blast crisis. 10-year OS was 10–20%. Busulfan/hydroxyurea and IFN-alpha were the prior standards.

BOARD TRAP

The dismal pre-TKI numbers are historical — modern TKI-era survival approaches that of the age-matched population.

5. TKI era 85% / >450,000

WHAT YOU SEE

85% survival, 450,000 prevalence in TKI era.

WHAT IT ENCODES

In the TKI era, ~85%+ achieve near-normal life expectancy and prevalence keeps rising (>450,000) because patients live on therapy. Annual mortality on TKI is only ~1–2%.

BOARD TRAP

Rising prevalence reflects survival, not rising incidence — incidence is stable.

6. Chronic phase <5% blasts

WHAT YOU SEE

Blue CHRONIC platform, 90%, <5% blasts.

WHAT IT ENCODES

~90% present in chronic phase: <10% blasts in blood/marrow (often cited <5%), controllable counts, minimal symptoms. This is the curable-into-deep-remission stage.

BOARD TRAP

Chronic phase by definition has <10% blasts; reaching ≥20% blasts defines blast crisis, an AML/ALL-like emergency.

7. Accelerated 60–70% 8-yr

WHAT YOU SEE

Orange ACCELERATED rainbow, 10–19% blasts, ≥20% basos, clonal evolution.

WHAT IT ENCODES

Accelerated phase: 10–19% blasts, ≥20% basophils, persistent thrombocytopenia or thrombocytosis, or clonal evolution (additional cytogenetic abnormalities). 8-year survival ~60–70%.

BOARD TRAP

≥20% BASOPHILS (not blasts) is an accelerated-phase criterion; ≥20% blasts jumps you to blast phase.

8. Blast phase ≥20% blasts

WHAT YOU SEE

Red BLASTIC explosion, ≥20% blasts, myeloid 60% / lymphoid 25%.

WHAT IT ENCODES

Blast crisis = ≥20% blasts in blood/marrow or extramedullary disease. ~60% myeloid, ~25% lymphoid. Median survival is months; treat the lineage (AML- or ALL-type chemo) plus a TKI, then transplant.

BOARD TRAP

Lymphoid blast crisis can mimic de novo Ph+ ALL — a prior chronic-phase history and basophilia point to CML blast crisis.

9. Cytogenetics mandatory

WHAT YOU SEE

Right wall: karyotype 9/22, MANDATORY banner.

WHAT IT ENCODES

Baseline marrow karyotype is mandatory to detect the Ph chromosome and additional clonal abnormalities (ACAs) that signal accelerated disease. The t(9;22) shuffles ABL1 from chr 9 to BCR on chr 22.

BOARD TRAP

Karyotype detects ACAs that FISH/PCR can miss — that is why baseline marrow cytogenetics is still required despite molecular testing.

10. FISH cannot detect ACAs

WHAT YOU SEE

FISH monitor, CANNOT DETECT ACAs note.

WHAT IT ENCODES

FISH confirms BCR::ABL1 fusion and is more sensitive than karyotype for the fusion itself, but it cannot detect additional chromosomal abnormalities outside the probed loci.

BOARD TRAP

FISH-positive does NOT replace karyotype — a normal FISH with new ACAs is a real progression scenario karyotype catches.

11. PCR monitoring

WHAT YOU SEE

PCR machine with waveform, step 3.

WHAT IT ENCODES

Quantitative RT-PCR for BCR::ABL1 transcript (International Scale) is the standard for monitoring response and minimal residual disease, far more sensitive than FISH or karyotype.

BOARD TRAP

PCR is for monitoring depth of response (MMR, MR4); diagnosis still needs cytogenetics/FISH to characterize the clone.

12. Survival/response scale

WHAT YOU SEE

Bottom bar: 100% diagnosis → PCyR ≤10% → CCyR ≤1% → MMR ≤0.1% → MR4 ≤0.01% → MR4.5 ≤0.0032%.

WHAT IT ENCODES

Response milestones on the International Scale: PCyR ~≤10%, CCyR ~≤1% (CCyR), MMR/MR3 ≤0.1%, MR4 ≤0.01%, MR4.5 ≤0.0032%. Deep molecular response (MR4/MR4.5) sustained is the prerequisite to attempt treatment-free remission.

BOARD TRAP

MMR (MR3, ≤0.1%) is NOT deep enough to stop a TKI — TFR requires sustained MR4–MR4.5.
